

36103 Statistical Thinking for Data Science

Assessment Task 1: Exploratory data analysis

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Executive summary

A telecommunication company ran a marketing campaign to promote a new subscription plan, but only a small group of customers signed up. The company wants to understand which customer segments respond the best, so that the future campaigns can be targeted more effectively.

This report explores 41,158 records of past campaign contacts using three exploratory techniques (univariate, bivariate, multivariate analysis) to identify which factors are associated with a successful subscription.

Overall, 11.3% of contacted customers subscribed. Students (31.4%) and retirees (25.1%) responded at roughly three times the average. The strongest single signal was prior campaign history, with customers who had subscribed a previous campaign at 65%, compared with roughly 14% among those whose previous campaign failed and 9% for clients never previously contacted. Contact method and timing matters, with cellular contact outperforms telephone contact and low number of contacts but high subscription rate in March, September, October and December.

The campaign's success was concentrated in a minority of customers. Focusing on previous responders and flexible-schedule segments, contacted by cellular, would be expected to lift conversion substantially compared with the current approach.

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Introduction

Problem statement

A telecommunication company recently launched a marketing campaign to promote the adoption of their new subscription plan among customers. The company wants to understand their customers and identify the customer segments that display the highest responsiveness to marketing campaigns.

Project aims and objectives

The business objective is to identify customers' characteristic demographic associated with higher subscription rate, which can be beneficial for future campaigns. The dataset contains 41,180 records of past campaign contacts, describing each customer through demographic attributes (age, job, marital status and education), financial attributes (existing loans, default history), campaign-contact attributes (contact method, month, day of week, number of contacts, days since previous contact) and prevailing macroeconomic indicators at the time of contact (employment variation rate, consumer price index, consumer confidence index, Euribor rate, number of employees). And finally, target variable (y) records whether the customer agreed to subscribe to a Telecom plan.

This report addresses the following questions, which will guide the exploratory analysis:

- RQ1: Which demographic and socioeconomic customer segments show the highest subscription rate?
- RQ2: How Does the outcome of a previous marketing campaign relate to a customer's likelihood of subscribing to the current offer?
- RQ3: Do the channel and timing of contact (method, time) relate to campaign success?
- RQ4: To what extent are prevailing macroeconomic conditions associated with subscription behaviour?

Based on an initial inspection of data, there are three working hypotheses:

- H1: Clients with more flexible personal circumstances (e.g. students and retirees) will show higher subscription rates than working-age customer segments
- H2: A successful outcome in a prior campaign will be one of the strongest predictors of a successful outcome in the current campaign.
- H3: Subscription rates will vary systematically with the macroeconomic climate at the time of the contact.

Methodology

Data preprocessing

Dataset Overview

The dataset consists of 41,180 rows and 21 columns with 10 numeric and 11 categorical variables. Prior to analysis, the raw CSV file has the following issues: the values within each row are separated by semicolons and some fields enclosed in quotation marks. During the data preprocessing stage, the values are therefore separated, and enclosing marks are removed from the beginning and end of each field.

The supplied dataset has some issues with information in the supplied data dictionary. The data dictionary references balance and day columns that do not exist in the CSV dataset, while the file contains a day_of_week column and no balance field.

Data quality issues and treatment

Issue	Description	Treatment
"Unknown" placeholders	Default (20.9%), education (4.2%), housing/loan (2.4%), job (0.8%), marital (0.2%) contain the value "unknown" which functions as missing data.	Retained as an explicit category for EDA, since the proportion in default is too large to safely drop.
Duplicate rows	12 fully duplicated records were found	12 duplicated rows were removed
Implausible age	10 records had age > 90 which is inconsistent with the rest of the distribution.	10 rows with age > 90 were removed
pdays placeholders	96.3% of records have pdays = 999, a sentinel value meaning "never previously contact"	Retained but interpreted as categorical variables (contacted vs not)
Target class imbalance	Only 11.3% of customers subscribed while 88.7% who did not	No resampling applied

After removing duplicate rows and implausible age, the dataset used for EDA contains 41,158 records.

Exploratory Data Analysis

There are three distinct EDA techniques which are applied to the dataset:

1. Univariate analysis: The technique is used to understand individual variables distribution.
2. Bivariate analysis: The technique is used to examine the relationship between independent variables and the target variable.
3. Multivariate analysis: The technique is used to assess how multiple variables are associated with each other and the target variable.

Univariate analysis

Figure 1 shows the distribution of the three numeric variables (age, call duration and number of contacts) through histograms. The first histogram illustrates that customer age is approximately bell-shaped and centred in the 30-45 range, with a long tail extending toward retirement age. The second chart indicates that the majority of calls was under 500 seconds (roughly 8.5 minutes), and call durations are heavily right-skewed. The last chart shows that the number of contacts per customer is also right-skewed, with most clients contacted only once or twice.

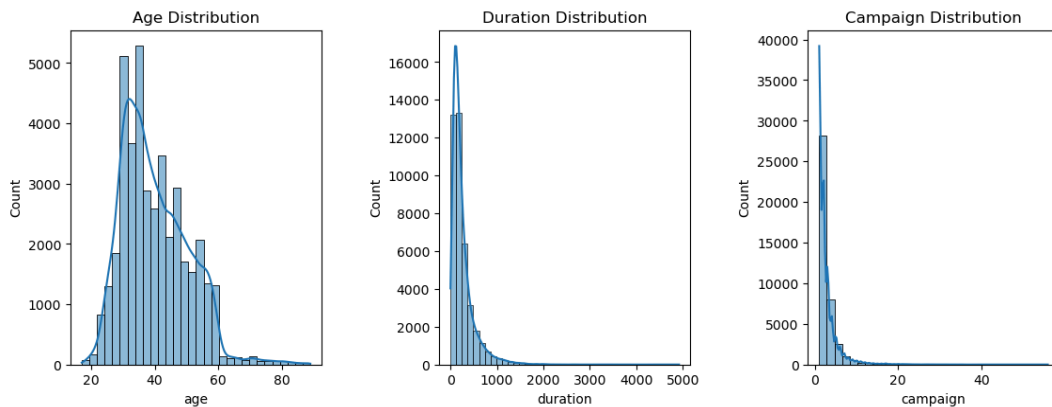


Figure 1. Distribution of age, call duration, and number of campaign contacts

The imbalance of subscription outcome was shown in Figure 2. The bar chart illustrates that 11.3% of clients in the dataset subscribed to the plan.

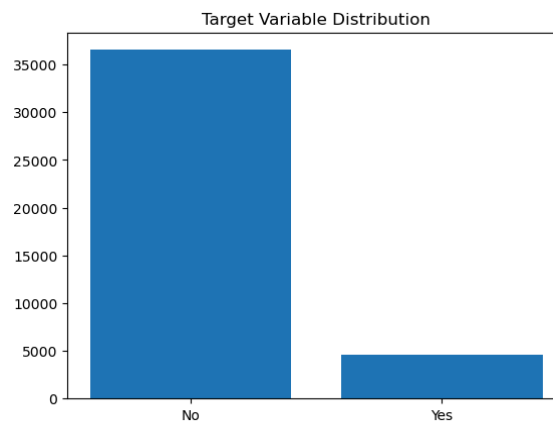


Figure 2. Target variable distribution

Figure 3 shows that the job distribution is highly uneven across categories. Admin and blue-collar workers account for the largest proportion of clients, followed by technicians. In contrast, students, unemployed individuals and housemaids represent relatively small groups.

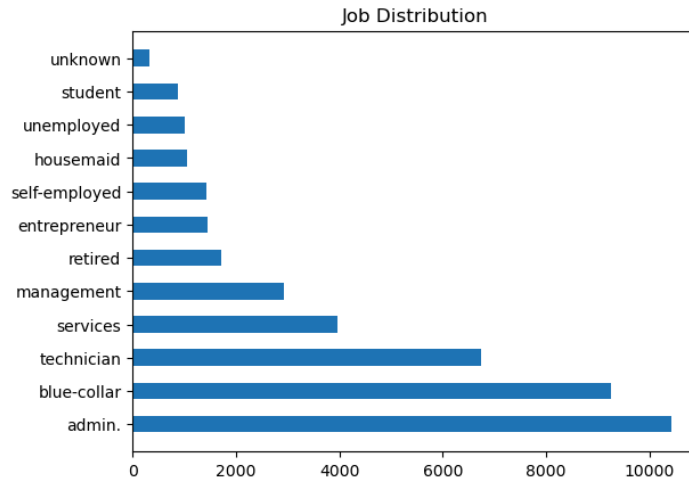


Figure 3. Job distribution

Bivariate analysis

To address RQ1 (Which demographic and socioeconomic customer segments show the highest subscription rate?), subscription rate was computed for each level of job and education variables (Figure 4). For jobs, students and retired customers show the highest subscription rates, at roughly three times the overall average (approximately 10%), while blue-collar and services workers account for the lowest rate. For education, the illiterate category shows the highest rate (22.2%), but this group contains only 18 customers, therefore this category should be treated as unreliable due to its very small sample size. Among the education levels, subscription rate rises broadly with education level: from basic education (7.8-10.2% with $n > 4000$ each) through high school and professional course (10.8-11.3%) to university degree, which combines a high rate at 13.7% with large sample size ($n = 12,160$).

This pattern is broadly consistent with hypothesis H1: customer segments with greater flexible times (students and retirees) or higher education level appear more likely to subscribe to the campaign.

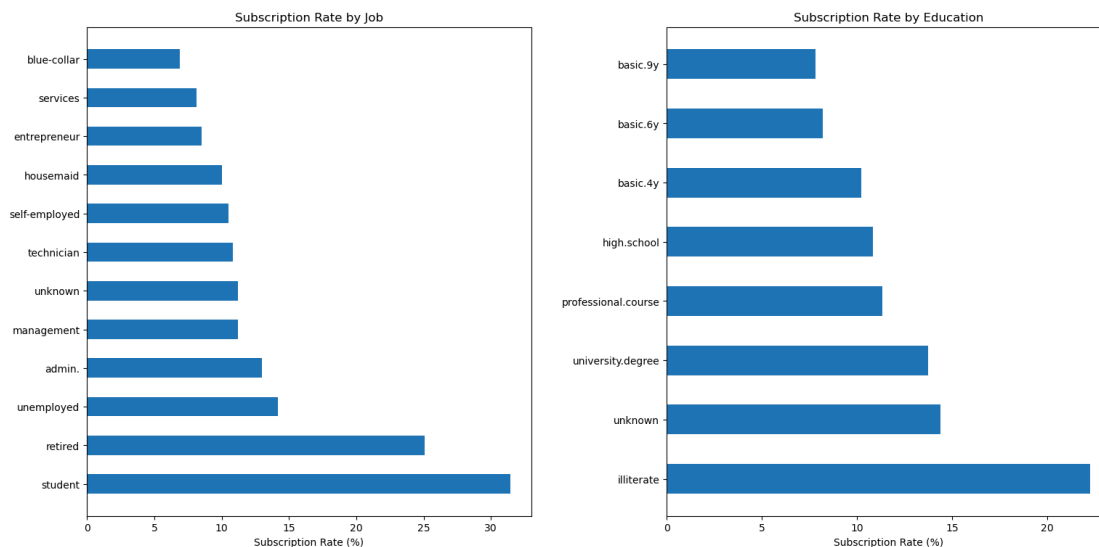


Figure 4. Subscription rates (%) by job, education

Addressing RQ2 (How Does the outcome of a previous marketing campaign relate to a customer's likelihood of subscribing to the current offer?). The left chart in Figure 5 shows that the outcome of clients' previous campaign was a success subscribed at approximately 65%, significantly higher than those whose previous campaign had failed (at roughly 14%) or clients who had never previously been contacted (at approximately 9%).

This bivariate relationship clearly supports hypothesis H2: A successful outcome in a prior campaign will be one of the strongest predictors of a successful outcome in the current campaign.

To address RQ3 (Do the channel and timing of contact (method, time) relate to campaign success?), contact (right chart in Figure 5) methods and campaign timing (Figure 6) were also examined. The contact method chart indicates that clients who were contacted by cellular phone account for a higher rate than those contacted by telephone. Figure 6 shows the number of contacts made each month against the monthly subscription rate: months such as March, September, October, December show relatively low numbers of contact but disproportionately high subscription rates, while mid-year months show high contact volume but low subscription rates.

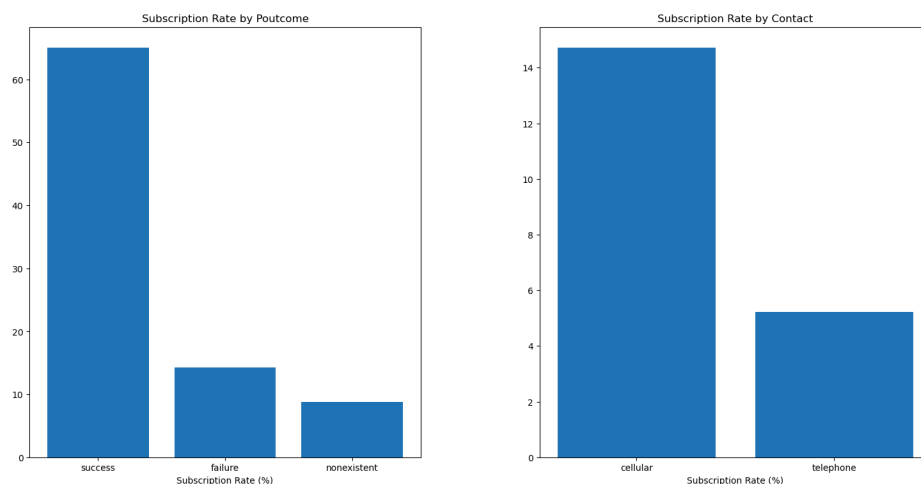


Figure 5. Subscription rates (%) by previous outcome, contact method

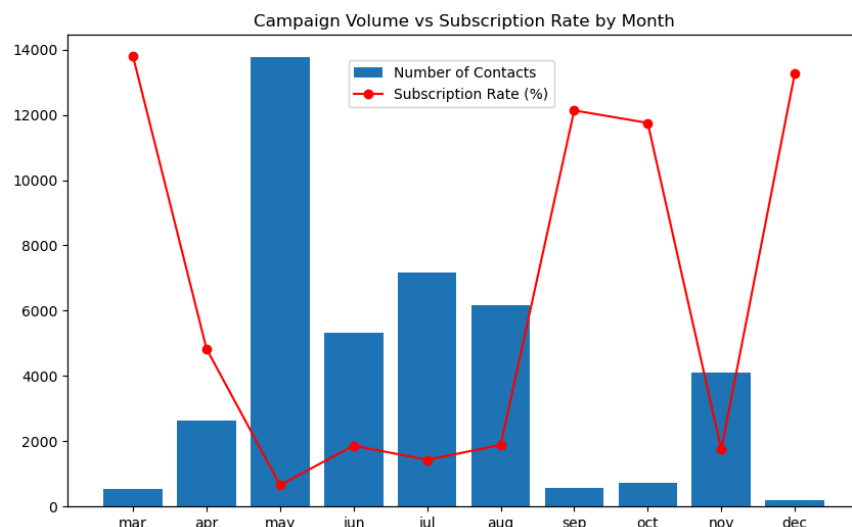


Figure 6. Number of contacts and subscription rate by month

Finally, Figure 7 compares the age distribution of clients who subscribed against those who did not. Customers who subscribed show a wider age range, which is also consistent with job-based finding that students and retired individuals are more responsive.

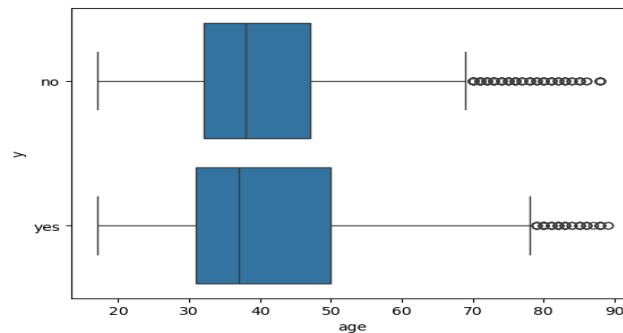


Figure 7. Age distribution by subscription outcome

Multivariate analysis

To address RQ4 (To what extent are prevailing macroeconomic conditions associated with subscription behaviour?) Figure 8 presents the correlation matrix for all numeric variables, including the target variable which was interpreted into a binary variable. Three macroeconomic indicators (employment variation rate, euribor 3 month rate, number employed) are themselves highly collinear ($r > 0.9$) and are negatively correlated with subscription (roughly -0.3 to -0.35), which support hypothesis H3 (Subscription rates will vary systematically with the macroeconomic climate at the time of the contact).

Besides, call duration shows the strongest correlation with the target variable ($r = 0.41$) but duration is only known once contact has concluded and is itself determined by outcome. Therefore, this variable must be excluded from any predictive model.

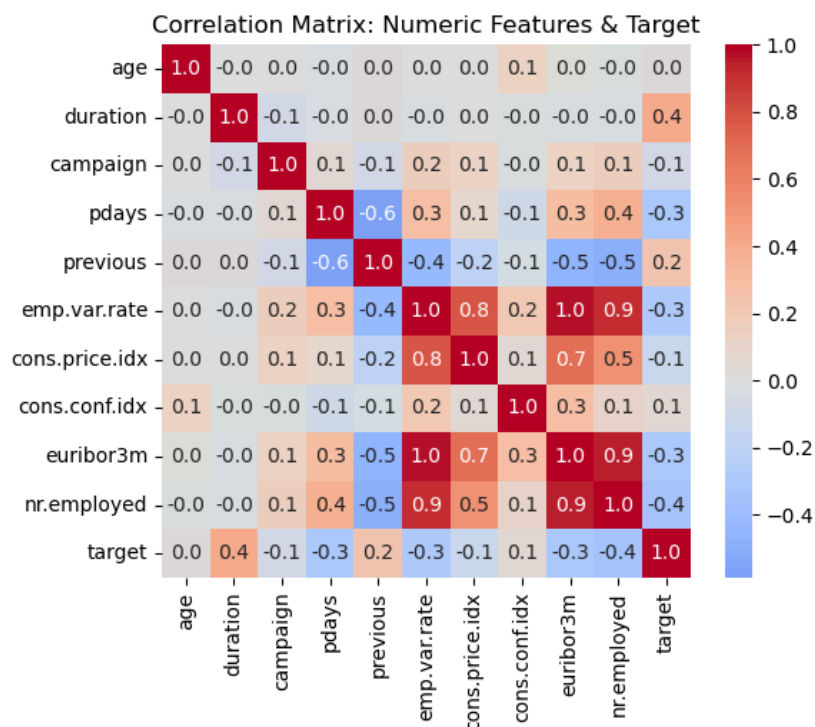


Figure 8. Correlation matrix of numeric features and the subscription target

Results

Key findings

Based on the analysis conducted, the following key insights were identified:

- The subscription rate in the campaign is strongly imbalanced (11.3% subscribed and 88.7% did not).
- Students and retirees have subscription rates roughly three times higher than the overall average rate and subscription rate rises with education level.
- Prior campaign outcome is the strongest predictor: customers with a previous success subscribe at approximately 65%, compared with roughly 14% among those whose previous campaign failed and 9% for clients never previously contacted.
- Cellular contact outperforms telephone contact. Besides, campaign timing matters, supported by low number of contacts but high subscription rate in March, September, October and December.
- Macroeconomic indicators are moderately negatively correlated with subscription and are highly collinear with one another ($r > 0.9$)